

Supporting Information for “Statistical inference combined with persistent homology for predicting fluid flow in porous media”

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Introduction. Figure S1 shows porous material structures amenable to persistent homology analysis, as suggested in Robins, Saadatfar, Delgado-Friedrichs, and Sheppard (2016); Figure S2, S3, and S4 show 0, 1, and 2 dimensional persistent diagrams for the Bentheimer sandstone network of Andrew, Bijeljic, and Blunt (2014), under nine simulated strain levels; Figure S5 shows average SSIM of the Rotleigend sandstone.

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Figure S1. Dimension 0 reveals the size of a pore and its connectivity. Region A corresponds to disconnected pores, such as the small pore on the lower left. Region B summarizes connected pores and pore throat radii. Dimension 1 identifies numbers of pores and grains. Region C reflects grains that contact each other. Region D shows non-convex pores. For example, when two grains are close to each other but not in contact, then the pore between them is non-convex. Region E displays non-convex grains, such as a broken ring-shaped pore encapsulated in a grain. Dimension 2 outlines the sizes of grains.

References

- Andrew, M., Bijeljic, B., & Blunt, M. J. (2014). Pore-scale imaging of trapped supercritical carbon dioxide in sandstones and carbonates. *International Journal of Greenhouse Gas Control*, 22, 1 - 14. Retrieved from <http://www.sciencedirect.com/science/article/pii/S1750583613004519> doi: <https://doi.org/10.1016/j.ijggc.2013.12.018>
- Robins, V., Saadatfar, M., Delgado-Friedrichs, O., & Sheppard, A. P. (2016). Percolating length scales from topological persistence analysis of micro-CT images of porous materials. *Water Resources Research*, 52, 315-329. doi: 10.1002/2015WR017937

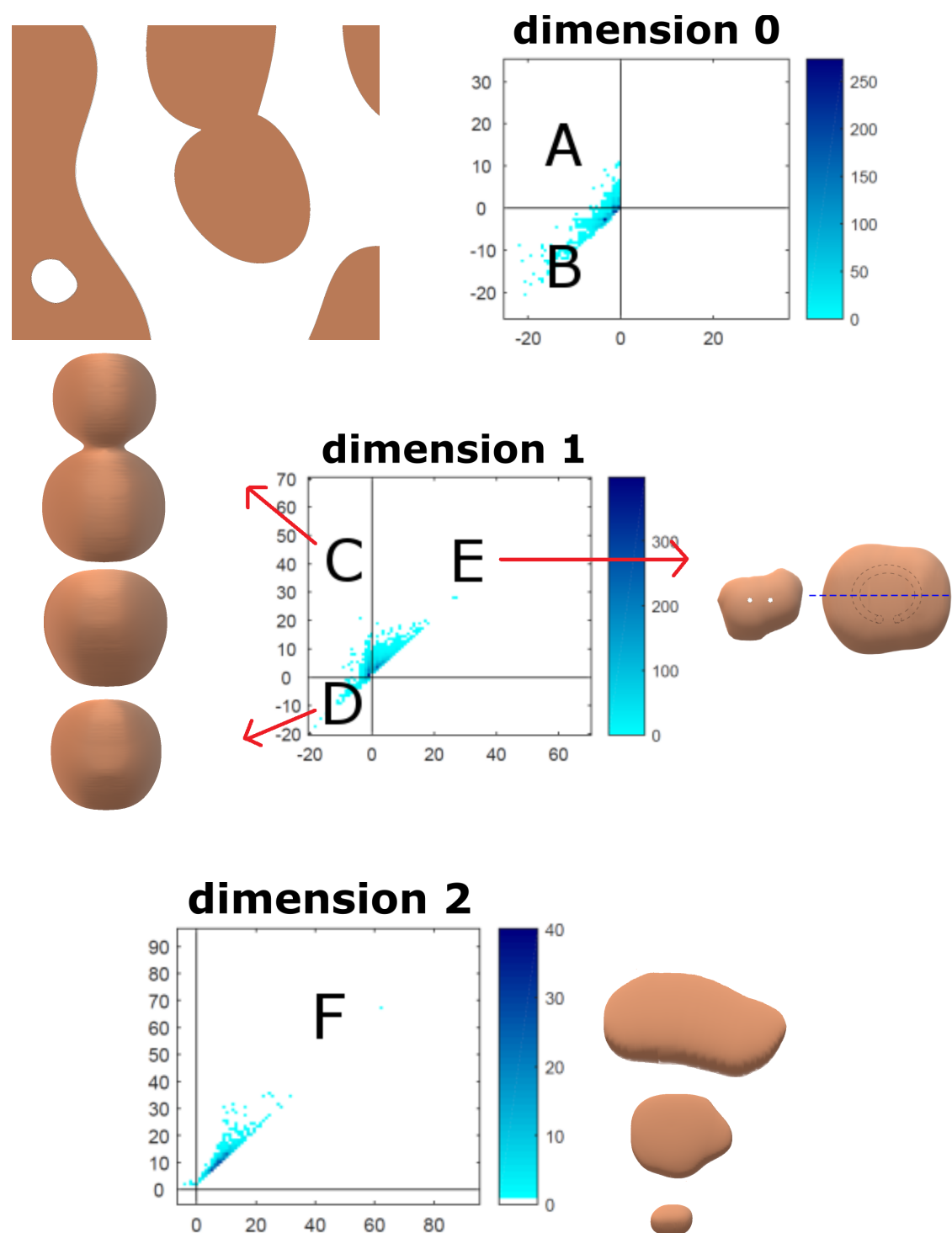


Figure S1. Example pore and grain structures for persistence diagrams by dimension.

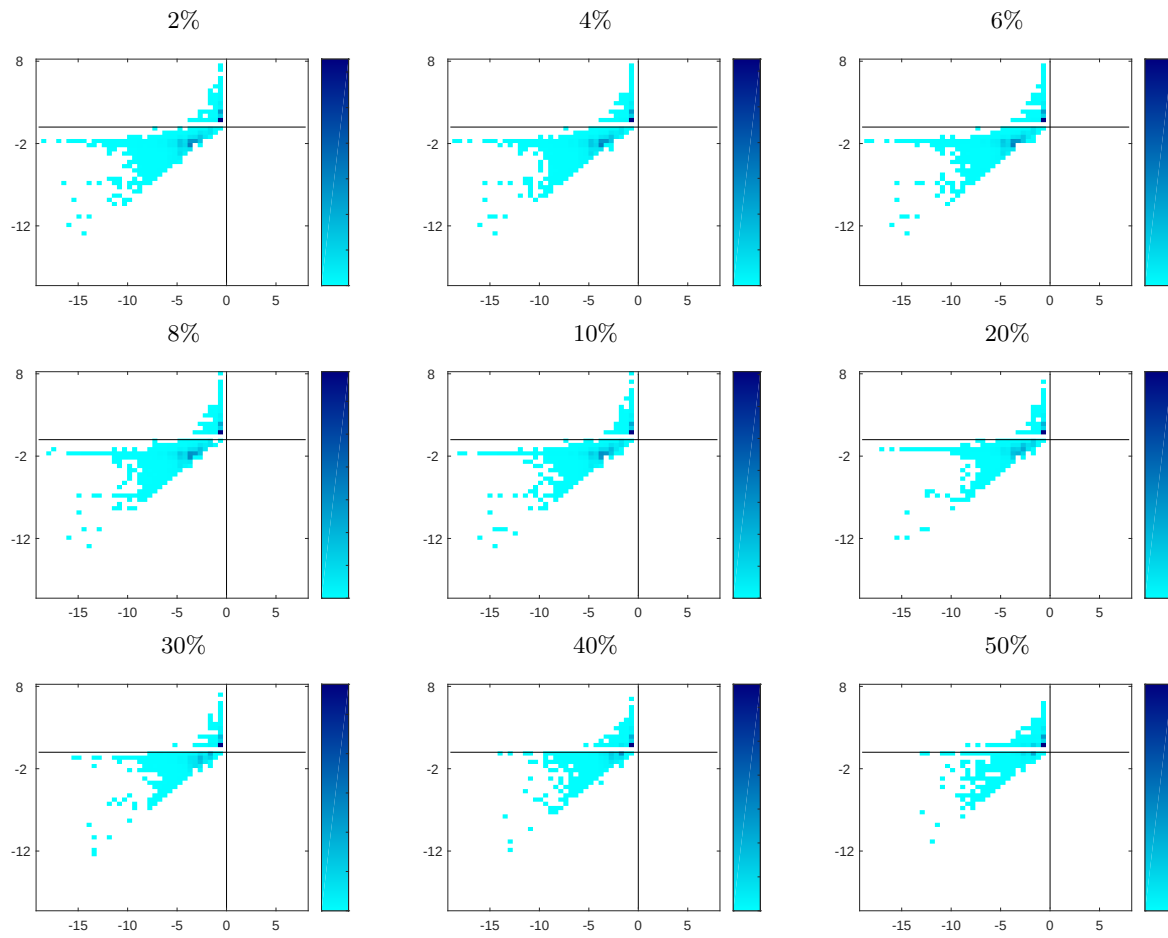


Figure S2. *Dimension 0* persistence diagrams under nine stress-strain levels: 2%–50%.

As stress-strain increases, connected-pore sizes and pore throat radii decrease (part B of Figure S1). Further, the number of disconnected pores increases (part A of Figure S1) and the number of connected pores decreases.

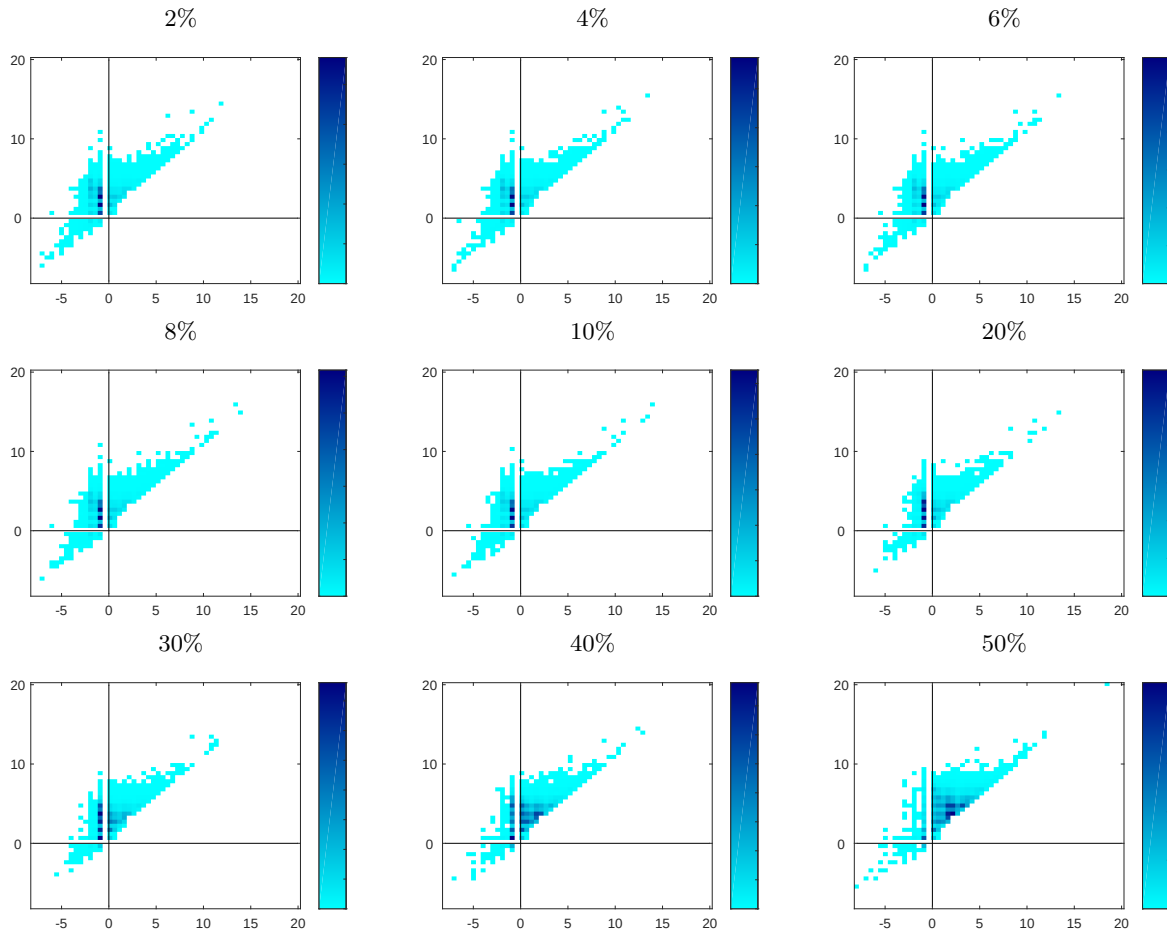


Figure S3. *Dimension 1* persistence diagrams under nine stress-strain levels: 2%–50%.

As stress-strain increases, contact grains (part C of Figure S1) glue together to form non-convex grains (part E of Figure S1).

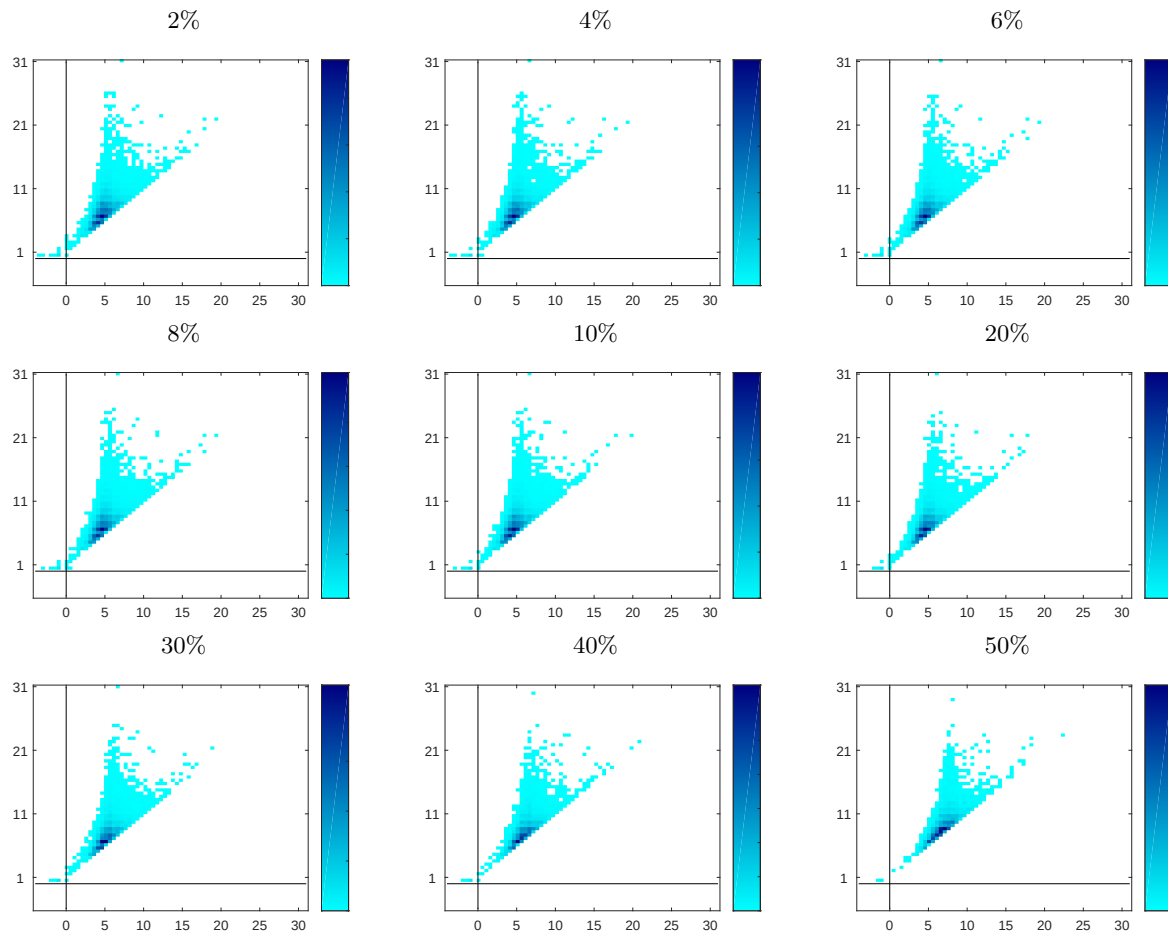


Figure S4. *Dimension 2* persistence diagrams under nine stress-strain levels: 2%–50%.

As stress-strain increases, the sizes of grains increases, illustrated by points migrating towards the upper-right.

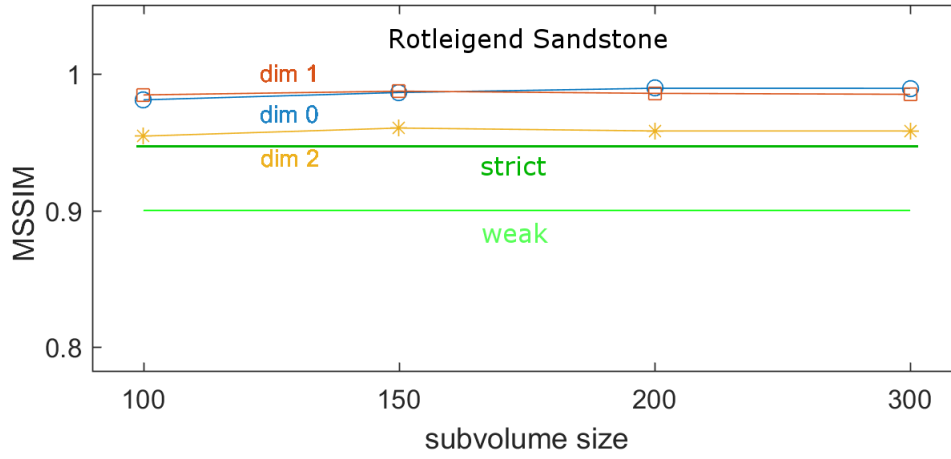


Figure S5. Average SSIM ($MSSIM_{PH}$) for Rotleigend sandstone. For all subsample sizes the SSIM exceeds both the weak (0.9) and strict (0.95) sREV thresholds. However, the sREV has not been reached. The data are merely uniformly bland because the physical sizes of subsamples are simply too small compared to the grain sizes. E.g., some subsamples lie inside a single grain.